

## **PDF PASSWORD CRACKING & PASSWORD SECURITY ASSESSMENT REPORT**

### **W3-PM — Password Cracking Lab CYBERSECURITY | NETWORKWALKS**

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|---------------------------|-------------------------------------------------------------------------------|
| <b>Report Information</b> | <b>Details</b>                                                                |
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| <b>Program / Batch</b>    | B083 – Networkwalks                                                           |
| <b>Module</b>             | Password Cracking / PDF Password Security                                     |
| <b>Date</b>               | 21 September 2026                                                             |
| <b>Tools Used</b>         | Kali Linux, pdf2john, John the Ripper, rockyou.txt, Networkwalks Online Tools |

#### **1. Liability Disclaimer**

I carried out these activities strictly on PDF files provided for the authorized educational cybersecurity laboratory and within the scope of the assigned Networkwalks practical.

The techniques demonstrated in this report are intended only for educational, ethical hacking, and authorized security-testing purposes. Password recovery or cracking against files, systems, accounts, or data without explicit authorization may be illegal and may result in legal, financial, professional, or other consequences.

The commands and techniques documented here should therefore only be reproduced in controlled laboratory environments, on personally owned systems, or where explicit permission has been granted. The tester is responsible for ensuring that all security-testing activities remain within the authorized scope.

This report documents the practical activity and its observed results; successful password recovery in a laboratory environment does not by itself indicate that the same technique will succeed against a properly protected real-world target.

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## 2. Introduction

This report documents a practical password-cracking exercise performed as part of the Networkwalks cybersecurity training program.

The objective of the activity was to understand how a password-protected PDF can be analyzed by first extracting its password hash and then attempting password recovery using a wordlist-based attack. The practical was performed in **Kali Linux** using **PDF2John** and **John the Ripper (JtR)**.

Three password-protected PDF files were used for the exercise. The workflow consisted of:

1. Extracting the password hashes from the PDF files using pdf2john.
2. Saving the extracted hashes into text files.
3. Using **John the Ripper** with the Kali Linux rockyou.txt wordlist.
4. Recovering the passwords successfully for the laboratory PDFs.
5. Verifying the recovered passwords by opening the corresponding protected PDF files.
6. Repeating the password-recovery process through the Networkwalks online password-cracking tools to compare the workflow with a faster web-based approach.

The exercise demonstrated the practical relationship between password-protected files, password hashes, wordlists, and dictionary-based password attacks. It also reinforced the importance of strong passwords and appropriate password-protection mechanisms.

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## 3. Tools Used

| Tool                                        | Purpose                                                                                      |
|---------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Kali Linux</b>                           | Operating system used for the password-cracking laboratory                                   |
| <b>pdf2john</b>                             | Extracted password-hash information from the protected PDF files                             |
| <b>John the Ripper (JtR)</b>                | Performed the password-recovery/dictionary attack against the extracted PDF hashes           |
| <b>rockyou.txt</b>                          | Wordlist used by John the Ripper to test candidate passwords                                 |
| <b>Networkwalks Online Hash Extractor</b>   | Extracted/processed the PDF password-hash information through the online interface           |
| <b>Networkwalks Online Password Cracker</b> | Performed password-recovery attempts through the web interface                               |
| <b>Firefox/Web Browser</b>                  | Used to access the Networkwalks online password-cracking tools and verify the protected PDFs |

The Kali Linux portion was performed locally in the controlled laboratory environment. The online portion was used as a second method of performing the same basic password-recovery task.

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## 4. Activities Performed

### 4.1 Password Hash Extraction and Cracking using Kali Linux

#### 4.1.1 Extracting the PDF Password Hash

The first step was to obtain the password-hash information from the protected PDF rather than attempting to directly modify or bypass the PDF protection.

I started by using the pdf2john command in the Kali Linux terminal.

For the first PDF, the command used was:

```
pdf2john My-Locked-PDF1.pdf > pdf_hash2.txt
```

This converted the password-protection information contained in the PDF into a format that could be processed by John the Ripper.

The extracted information was stored in:

```
pdf_hash2.txt
```

The same process was subsequently performed for the other protected PDF files, with their extracted hashes saved into separate text files.

For example, the second PDF was processed using:

```
pdf2john My-Locked-PDF2.pdf > pdf_hash3.txt
```

The terminal output shown in the evidence confirmed that John the Ripper recognized the resulting entries as PDF password hashes, including the supported PDF/MD5/SHA2/RC4/AES format.

This separation between **hash extraction** and **password cracking** was an important part of the exercise: pdf2john prepared the data, while John the Ripper performed the password-recovery attempt.

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#### 4.1.2 Using John the Ripper with the Kali Wordlist

After extracting the hash, I used **John the Ripper** with the rockyou.txt wordlist available in Kali Linux.

The command used was:

```
john --wordlist=/usr/share/wordlists/rockyou.txt pdf_hash2.txt
```

Initially, the command was executed from a different directory and John reported that the hash file could not be found:

```
stat: pdf_hash.txt: No such file or directory
```

I then changed to the directory containing the extracted hash:

```
cd Downloads
```

and ran the John the Ripper command again.

John successfully loaded the PDF hash and began testing password candidates from the wordlist.

The terminal evidence shows John reporting:

Loaded 1 password hash

and subsequently completing the cracking process.

The successful result was then verified by opening the corresponding protected PDF and entering the recovered password. The PDF opened successfully and displayed the Networkwalks laboratory flag, confirming that the recovered password was valid.

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#### **4.1.3 Cracking the Remaining PDF Files**

The same workflow was repeated for the remaining protected PDFs:

##### **Step 1 — Extract the hash**

```
pdf2john <PDF-file> > <hash-file>
```

##### **Step 2 — Run John the Ripper**

```
john --wordlist=/usr/share/wordlists/rockyou.txt <hash-file>
```

##### **Step 3 — Wait for the wordlist attack to complete.**

##### **Step 4 — Verify the recovered password against the original protected PDF.**

The evidence shows successful password recovery for the laboratory PDFs. For example, one of the terminal sessions shows John recovering the password for My-Locked-PDF2.pdf and completing the session successfully.

The recovered passwords were then tested against their respective PDF files. The successful opening of the PDFs provided an independent verification that the recovered credentials were correct.

The practical therefore demonstrated the complete local workflow:

Protected PDF

↓

pdf2john

↓

Extracted PDF Hash

↓

John the Ripper

↓

rockyou.txt Wordlist

↓

Recovered Password



Open / Verify PDF

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## 4.2 Password Recovery Using Networkwalks Online Tools

After completing the local Kali Linux workflow, I performed the same general exercise using the Networkwalks online password-cracking interface.

The online workflow was comparatively faster and required fewer local commands.

The process involved:

1. Providing the PDF/hash information to the online hash-extraction interface.
2. Obtaining the corresponding PDF hash.
3. Providing the hash to the online password-cracking tool.
4. Running the built-in password list against the hash.
5. Observing the recovered password when a match was found.
6. Using the recovered password to open and verify the corresponding PDF.

The screenshots show the online tool testing candidate passwords and displaying a successful **"PASSWORD CRACKED SUCCESSFULLY"** result when a matching password was found.

For example, one of the online attempts displayed:

Tried: 35 / 100

MATCH: 1qaz2wsx

The interface subsequently displayed the successful password-recovery result.

Another online attempt progressed through the supplied password candidates until a matching password was identified. The recovered passwords were then used to verify access to the respective laboratory PDFs.

Compared with the Kali terminal workflow, the online method abstracted much of the underlying process. Instead of manually executing pdf2john, preparing hash files, selecting the wordlist, and running John, the web interface combined several of these steps into a simplified workflow.

This was useful for comparison because it demonstrated that the underlying concept remains the same:

PDF Password Protection



Hash Extraction



Password Candidates / Wordlist



Hash Comparison



Matching Password

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## 5. Summary of Results

The password-cracking laboratory was completed successfully for the three assigned protected PDF files.

| Activity                              | Result                 |
|---------------------------------------|------------------------|
| PDF hash extraction using pdf2john    | Successfully completed |
| John the Ripper installation/use      | Successfully completed |
| Kali rockyou.txt wordlist             | Successfully used      |
| Password recovery for PDF 1           | Successful             |
| Password recovery for PDF 2           | Successful             |
| Password recovery for PDF 3           | Successful             |
| Password verification by opening PDFs | Successful             |
| Networkwalks online hash extraction   | Successfully completed |
| Networkwalks online password cracking | Successfully completed |
| Laboratory flags                      | Successfully captured  |

| PDF File           | Password Recovered | Verification                     |
|--------------------|--------------------|----------------------------------|
| My-Locked-PDF1.pdf | password1          | Successfully opened and verified |
| My-Locked-PDF2.pdf | password1          | Successfully opened and verified |
| My-Locked-PDF3.pdf | 1qaz2wsx           | Successfully opened and verified |

The screenshots provide evidence of both the local Kali Linux workflow and the online workflow. The local terminal evidence demonstrates hash extraction, John the Ripper execution, wordlist processing, and successful completion. The PDF screenshots demonstrate that the recovered passwords were valid because the protected files could subsequently be opened.

One practical observation was that the password-recovery time depended heavily on where the correct password appeared within the candidate wordlist. John the Ripper was able to process large numbers of candidates per second in the laboratory environment, while the online tool presented a simpler interface for the same general task.

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## **6. Recommendations Against This Type of Attack in Real-World Executions**

The laboratory demonstrates why weak or commonly used passwords can present a security risk. Organizations and individuals should take several precautions when protecting sensitive PDF documents and other files.

### **6.1 Use Strong, Unique Passwords**

Passwords should not be based on common words, predictable sequences, names, simple patterns, or passwords commonly found in public wordlists.

A strong password should have sufficient length and unpredictability and should not be reused across different systems or files.

### **6.2 Avoid Dictionary-Based Passwords**

The exercise specifically demonstrated the effectiveness of a dictionary/wordlist attack against passwords contained in a common password list.

Passwords that appear in commonly available wordlists can potentially be recovered very quickly when the underlying protection permits efficient password verification.

### **6.3 Use Long Passphrases**

For manually managed PDF passwords, a long, unique passphrase can provide substantially more resistance to dictionary-based guessing than a short common password.

### **6.4 Do Not Reuse Passwords**

A password used for a PDF should not also be used for email, cloud storage, administrative accounts, or other services. Compromise of one password should not automatically expose other resources.

### **6.5 Protect Passwords Separately**

When sensitive documents are transferred, the document and its password should not be sent through the same insecure communication channel.

### **6.6 Use Appropriate Encryption**

For sensitive information, organizations should use modern, appropriately configured encryption mechanisms rather than relying solely on weak or legacy password protection.

### **6.7 Perform Authorized Security Testing**

Security teams should periodically test their organization's password policies and protected documents using authorized security assessments. As emphasized in the previous Networkwalks report, security testing should remain within a formally authorized scope.

### **6.8 Treat Successful Password Recovery as a Security Finding**

If an authorized security assessment successfully recovers a password from a protected document, the finding should be documented and investigated. The organization should determine whether the password was weak, reused, exposed in a wordlist, or protected by an unsuitable encryption configuration.

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## **7. Conclusion**

This practical provided hands-on experience with PDF password recovery using both **Kali Linux** and Networkwalks' online security tools.

The local exercise began with extracting the password-protection information from the PDFs using pdf2john. The resulting hash files were then processed with **John the Ripper**, using the rockyou.txt wordlist available in Kali Linux. The passwords for all three laboratory PDFs were successfully recovered and verified by opening the protected files.

The second part of the exercise used the Networkwalks online hash-extraction and password-cracking tools. This provided a faster and more simplified interface for performing the same general password-recovery process.

The practical demonstrated that password security depends not only on having password protection enabled, but also on the strength and unpredictability of the password and the underlying protection mechanism. A weak password that appears in a commonly available wordlist can be recovered through a dictionary-based attack.

Overall, the exercise improved my practical understanding of **PDF hash extraction, wordlist-based password attacks, John the Ripper, password verification, and the security implications of weak passwords**. It also reinforced the importance of conducting password-cracking activities only within an authorized laboratory or security-testing scope, consistent with the approach used throughout the previous Networkwalks report.

## 8. Evidences Collected



